Business Requirements Document (BRD) — Courier & Logistics DW

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Project Title: Enterprise Courier & Logistics Data Warehouse & BI Solution (LogiTrack DWBI)
Module: DL3004 Data Warehousing & Business Intelligence
Phase: Phase 1 (30% Weightage)
Methodology: Ralph Kimball Dimensional Modeling Lifecycle

1. Document Control

1.1 Document Information

This Business Requirements Document (BRD) establishes the definitive business, functional, and governance requirements for the implementation of the LogiTrack Enterprise Data Warehouse. It serves as the baseline agreement between business stakeholders and the data warehouse architecture team for Phase 1 of module DL3004.

1.2 Revision History

Version	Date	Author	Summary of Changes
0.1	October 24, 2026	Taha Imran	Initial requirements capture and stakeholder interview synthesis.
0.2	October 25, 2026	Taha Imran	Incorporated Kimball fact archetype rules (pure factless assignment table).
1.0	October 25, 2026	Taha Imran	Finalized BRD with full KPI formulas, RTM, Acceptance Criteria, and Viva Defense.

2. Business Context & Strategic Objectives

LogiTrack operates an extensive hub-and-spoke last-mile delivery network across multiple metropolitan and secondary commercial regions. Operations incorporate regional sorting hubs, last-mile delivery routes, and a mixed fleet of full-time and contractual delivery riders serving business-to-business (B2B) corporate contracts and business-to-consumer (B2C) e-commerce packages.

2.1 Operational Pain Points & Baseline Metrics

The business faces critical visibility and operational bottlenecks (*assumed baseline figures for this case study*):

- **Elevated First-Attempt Delivery Failures:** Fluctuating between 18% and 24%, triggering expensive re-delivery attempts and customer SLA penalties.
- **Unmonitored Hub Backlogs & Dwell Spikes:** Evening parcel accumulation at hubs creates sorting congestion, extending transit dwell times past 14 hours.
- **Imbalanced Fleet Utilization:** High-density urban corridors overburden riders with excessive shifts, while peripheral routes experience idle labor.
- **Unmeasured Route Profitability:** Operating costs (fuel, labor, re-delivery attempt penalties) are disconnected from customer billing rate cards.

2.2 Core Warehouse Objectives

1. Deploy a centralized enterprise dimensional data warehouse using the **Kimball Lifecycle Methodology**.
2. Model three core business process fact archetypes: atomic delivery attempt transactions (`Fact_Delivery_Attempt`), daily hub backlog snapshots (`Fact_Daily_Hub_Backlog`), and pure scheduled workforce route coverage (`Fact_Rider_Route_Assignment`).
3. Provide an integrated bus architecture of conformed enterprise dimensions to eliminate reporting silos.

3. Stakeholder Analysis

The warehouse serves five key organizational personas across executive, logistical, and commercial operations.

Stakeholder Persona	Organizational Role	Core Analytical Needs & Pain Points	Primary DW Use Case	Access Pattern
Chief Operating Officer (COO)	Executive directing network SLA compliance, cost control, and regional scaling.	Lacks holistic visibility into network-wide on-time performance and regional SLA penalties. Cannot compare hub efficiency benchmarks.	Monitors network OTDR, tracks regional transit velocity, and allocates capital investment for line-haul transport.	Executive BI Dashboard (Monthly/Quarterly roll-up; regional drill-down).
Hub / Logistics Center Manager	Operational manager directing sorting shift labor, inbound offloading, and dispatch.	Blind to cumulative end-of-day backlogs and parcel dwell-time distributions. Staffing shifts are reactive rather than predictive.	Identifies bottlenecked transit lanes, analyzes backlog aging trends, and schedules sorting floor labor shifts.	Daily Operational Snapshot Report (Scheduled at 06:00 AM daily).

Fleet & Dispatch Supervisor	Field supervisor responsible for daily route coverage, driver dispatch, and balance.	Route assignments managed manually on spreadsheets. Cannot evaluate first-attempt delivery success vs. driver route familiarity.	Identifies unassigned route coverage gaps, audits first-attempt failure rates, and tracks rider productivity.	Weekly Fleet Performance Matrix (Drill-across assignment and delivery facts).
CFO / Commercial Director	Financial executive overseeing customer billing contracts, cost accounting, and profit.	Cost-to-serve (labor, fuel, re-attempt penalties) is disconnected from commercial billing. Unprofitable B2B accounts remain undetected.	Analyzes route-level gross margins, evaluates corporate customer contract profitability, and models tiered service pricing.	Monthly Customer & Route Profitability OLAP Cube (Slice-and-dice).
Customer Support Lead	Tactical lead responsible for resolving delivery exceptions, recipient claims, and CSAT.	Receives customer complaints with no centralized visibility into recurring failure reason codes (e.g., recipient absent, bad address).	Analyzes delivery exception patterns, identifies chronic delivery failure hot-spots, and refines address validation workflows.	Ad-hoc Exception Explorer & CSAT Reports (Filter by status, failure code).

4. Business Requirements

Business requirements define the analytical capabilities required by stakeholders, prioritized using the **MoSCoW** framework (Must, Should, Could, Won't).

BR ID	Business Requirement Description	Primary Stakeholder	Priority	Process Area
BR-01	The warehouse must measure On-Time Delivery Rate (OTDR) against established customer service level agreements (SLAs) across all service tiers, hubs, and regions.	COO	**Must**	Delivery Attempts (Transaction)
BR-02	The warehouse must capture the outcome of every delivery attempt, classifying successes, failures, and granular failure exception reason codes.	Customer Support Lead / Dispatch Supervisor	**Must**	Delivery Attempts (Transaction)

BR-03	The warehouse must monitor daily end-of-day parcel backlog counts, inflow/outflow volumes, and parcel dwell-time aging at each sorting hub.	Hub Logistics Manager	**Must**	Hub Backlog (Periodic Snapshot)
BR-04	The warehouse must record daily scheduled rider-to-route coverage rosters to track fleet deployment independent of whether deliveries occurred.	Fleet & Dispatch Supervisor	**Must**	Fleet Allocation (Factless)
BR-05	The warehouse must compute rider delivery productivity (Deliveries per Rider-Hour) by cross-referencing completed deliveries against scheduled shift durations.	Fleet & Dispatch Supervisor	**Must**	Multi-Process Drill-Across
BR-06	The warehouse must calculate the net contribution margin and cost-to-serve per delivery route by aggregating delivery revenues against estimated labor and fuel costs.	CFO / Commercial Director	**Must**	Delivery Attempts (Financial)
BR-07	The warehouse should evaluate corporate customer account profitability, order volume trends, and SLA compliance to support contract renegotiations.	CFO / Commercial Director	**Should**	Delivery Attempts (Customer Analytics)
BR-08	The warehouse should preserve the historical lineage of rider home hub reassignments and operational status changes over time.	Fleet & Dispatch Supervisor	**Should**	Workforce Master Data (SCD)
BR-09	The warehouse should maintain a complete history of customer corporate tier changes, credit limits, and delivery address relocations.	Commercial Director	**Should**	Customer Master Data (SCD)
BR-10	The warehouse could analyze the impact of physical parcel characteristics (weight, volumetric class) on vehicle capacity utilization and delivery success.	Hub Manager / COO	**Could**	Cross-Process Analysis

5. Functional Requirements for the Data Warehouse

Functional requirements specify the schema structures, dimensional models, and Kimball techniques required to implement the business requirements.

FR ID	Functional Warehouse Requirement	Linked BR	Supporting Schema Architecture / Kimball Technique
FR-01	Implement an atomic transaction fact table (`Fact_Delivery_Attempt`) where each record represents a single physical delivery attempt event.	BR-01, BR-02, BR-06	**Transaction Fact Table:** Grain is one row per delivery attempt. Contains additive measures for delivery fees, labor cost, fuel cost, and attempt duration.
FR-02	Implement a daily periodic snapshot fact table (`Fact_Daily_Hub_Backlog`) capturing end-of-day parcel accumulation at each hub.	BR-03	**Periodic Snapshot Fact Table:** Grain is one row per hub, date, parcel type, and backlog status at 23:59:59 daily. Contains semi-additive inventory and dwell measures.
FR-03	Implement a pure factless fact table (`Fact_Rider_Route_Assignment`) storing scheduled workforce coverage rosters with zero numeric measures.	BR-04	**Factless Fact Table:** Grain is one row per rider assigned to a route, hub, shift, and date. Consists entirely of surrogate foreign keys.
FR-04	Create a conformed Shift Dimension (`Dim_Shift`) containing standard shift hours and scheduling attributes to enable derived productivity metrics.	BR-05	**Conformed Dimension:** Contains `Standard_Shift_Hours` (e.g., 8.0 hrs), allowing the BI semantic layer to compute Deliveries per Rider-Hour without corrupting factless grain.
FR-05	Architect conformed Date, Hub, and Route/Region dimensions shared across all three fact tables to enable drill-across queries.	BR-01 to BR-10	**Conformed Dimensions (Bus Architecture):** Standardized surrogate keys across Date, Hub, and Route guarantee mathematically sound drill-across unions.
FR-06	Implement Slowly Changing Dimension (SCD) Type 2 tracking for `Dim_Rider` to preserve historical performance when a rider transfers hubs.	BR-08	**SCD Type 2 Dimension:** Attributes include `Current_Row_Indicator`, `Effective_Date`, and `Expiration_Date` on home hub and vehicle class changes.
FR-07	Implement SCD Type 2 tracking for `Dim_Customer` to maintain accurate historical attribution across corporate account tier and address changes.	BR-09	**SCD Type 2 Dimension:** Captures customer tier promotions/demotions and billing address shifts without overwriting past transactional context.
FR-08	Implement a consolidated Delivery Status Junk Dimension (`Dim_Delivery_Status`) combining	BR-02	**Junk Dimension:** Eliminates operational flags and text indicators from the fact table,

	attempt outcome, failure code, and delivery signature requirement.		suppressing snowflake normalization and optimizing columnar compression.
FR-09	Incorporate fully additive financial measures into `Fact_Delivery_Attempt` (`Delivery_Fee_Amount`, `Estimated_Labor_Cost`, `Estimated_Fuel_Cost`, `Net_Margin_Amount`).	BR-06, BR-07	**Fully Additive Fact Measures:** Stored as decimal currency fields that can be sliced, diced, and rolled up across any dimensional hierarchy.
FR-10	Pre-compute dwell-time aging buckets and backlog classification flags in `Fact_Daily_Hub_Backlog`.	BR-03	**Semi-Additive Measures:** `Backlog_Parcel_Count` and `Avg_Dwell_Hours` sum across hubs and parcel types, but cannot be added across time (must be averaged).
FR-11	Build a comprehensive Date Dimension (`Dim_Date`) supporting standard calendar, fiscal calendar, day-of-week, weekend, and statutory holiday hierarchies.	BR-01, BR-03, BR-04	**Role-Playing / Conformed Dimension:** Supports discrete date roles (Attempt Date, Scheduled Delivery Date, Snapshot Date, Assignment Date).
FR-12	Support BI semantic layer drill-across queries linking `Fact_Rider_Route_Assignment` and `Fact_Delivery_Attempt` via common conformed keys.	BR-05	**Multi-Fact Co-Processing:** Query engine matches facts along conformed Date, Rider, and Route keys, applying `Dim_Shift.Standard_Hours` as the divisor.

6. Key Performance Indicator (KPI) Definitions

This section provides technical definitions for all five primary organizational KPIs. All target figures are **assumed baseline targets for this case study**.

1. On-Time Delivery Rate (OTDR)

Business Definition: Percentage of shipments successfully completed within customer SLA commitment windows.

```
OTDR = (Count of Successful Attempts Where Attempt_Timestamp <= SLA_Deadline / Total Successful Delivery Attempts) * 100
```

Fact Grain: One row per physical delivery attempt in `Fact_Delivery_Attempt`.
Source Facts & Dimensions: `Fact_Delivery_Attempt`, `Dim_Date`, `Dim_Service_Type`, `Dim_Hub`, `Dim_Route`.

Measure Type: Fully Additive Counts; Derived Non-Additive Ratio (%).
Target Benchmark: >= 95.0% (*Assumed baseline target for this case study; current baseline 82%*).
Metric Owner: Chief Operating Officer (COO).

2. First-Attempt Delivery Rate (FADR)

Business Definition: Percentage of packages successfully delivered on the initial attempt, avoiding secondary re-delivery runs.

```
FADR = (Count of Successful Deliveries Where Attempt_Sequence_Number = 1 / Total Distinct Shipments Attempted) * 100
```

Fact Grain: One row per physical delivery attempt in `Fact_Delivery_Attempt`.
Source Facts & Dimensions: `Fact_Delivery_Attempt`, `Dim_Date`, `Dim_Delivery_Status`, `Dim_Customer`, `Dim_Route`.
Measure Type: Fully Additive Counts; Derived Non-Additive Ratio (%).
Target Benchmark: >= 85.0% (*Assumed baseline target for this case study; current baseline 76%*).
Metric Owner: Fleet & Dispatch Supervisor / Customer Support Lead.

3. Hub Backlog Ratio (HBR) & Backlog Dwell Index

Business Definition: Proportion of parcels staged at a sorting hub at End-of-Day relative to total daily inbound volume, alongside average dwell latency.

```
HBR = (End_of_Day_Backlog_Parcel_Count / Total_Daily_Inbound_Parcel_Count) * 100 Backlog Dwell Index = Average Dwell Time of Backlogged Parcels (in hours)
```

Fact Grain: One row per hub, date, parcel type, and backlog status at 23:59:59 daily.
Source Facts & Dimensions: `Fact_Daily_Hub_Backlog`, `Dim_Date`, `Dim_Hub`, `Dim_Parcel_Type`.
Measure Type: Semi-Additive Fact Counts (sum across hubs, average across dates); Non-Additive Ratio.
Target Benchmark: HBR <= 8.0%; Average Dwell <= 6.0 Hours (*Assumed baseline targets for this case study*).
Metric Owner: Hub / Logistics Center Manager.

4. Deliveries per Rider-Hour (DPRH)

Business Definition: Average delivery productivity achieved per scheduled rider working hour across designated routes and shifts.

```
DPRH = (Total Successful Deliveries from Fact_Delivery_Attempt) / Sum(Dim_Shift.Standard_Shift_Hours from matching records in Fact_Rider_Route_Assignment)
```

Fact Grain: Multi-fact drill-across joining Attempt-grain with Assignment-grain along conformed `Date_Key`, `Rider_Key`, `Route_Key`.
Source Facts & Dimensions: `Fact_Delivery_Attempt`, `Fact_Rider_Route_Assignment`, `Dim_Rider`, `Dim_Shift`, `Dim_Route`, `Dim_Date`.
Measure Type: Derived Cross-Process Metric (Ratio of transactional delivery count to scheduled shift hours).
Target Benchmark: >= 3.5 Deliveries / Hour (*Assumed baseline target for this case study*).
Metric Owner: Fleet & Dispatch Supervisor.

5. Route Net Contribution Margin (RNCM)

Business Definition: Net financial margin generated by a specific delivery route corridor after deducting operating labor and fuel expenses.

```
RNCM = Sum(Delivery_Fee_Amount) - Sum(Estimated_Labor_Cost + Estimated_Fuel_Cost) Margin % =
(RNCM / Sum(Delivery_Fee_Amount)) * 100
```

Fact Grain: One row per physical delivery attempt in `Fact_Delivery_Attempt`.
Source Facts & Dimensions: `Fact_Delivery_Attempt`, `Dim_Route`, `Dim_Hub`, `Dim_Service_Type`, `Dim_Customer`, `Dim_Date`.
Measure Type: Fully Additive Currency Amounts ($); Derived Percentage (%).
Target Benchmark: >= 22.0% Net Contribution Margin (*Assumed baseline target for this case study*).
Metric Owner: Chief Financial Officer (CFO) / Commercial Director.

7. Data Requirements & Source System Architecture

The warehouse integrates operational data from four core transactional source systems to construct dimensional data marts.

Source System	Technology Stack	Extracted Entities & Business Keys	Volume Profile	Extract Cadence & Target
Mobile Dispatch System (MDS)	PostgreSQL / Mobile REST API	Delivery attempt logs, physical attempt timestamps, GPS coordinates, recipient signature flags, exception failure codes (`Attempt_ID`, `Tracking_No`).	~15,000 – 25,000 records daily.	Hourly micro-batch delta scraping. → `Fact_Delivery_Attempt`, `Dim_Delivery_Status`

Hub Warehouse Management (WMS)	Oracle / Barcode Laser Scanners	Inbound scan timestamps, sorting bin assignments, outbound line-haul dispatch scans, end-of-day staging inventory logs.	~30,000 barcode scan events daily.	Daily batch at 00:30 AM via staging view. → `Fact_Daily_Hub_Backlog`, `Dim_Hub`
Commercial Billing & ERP	Microsoft Dynamics / SQL Server	Corporate customer accounts, billing rate cards, service tier agreements, delivery fees, fuel surcharge adjustments, cost allocations.	~10,000 invoices daily; 1,500 active customer records.	Daily batch at 02:00 AM via CDC. → `Fact_Delivery_Attempt` (financials), `Dim_Customer`
Workforce & Fleet Management	MySQL Web App / CSV Rosters	Scheduled rider shift rosters, assigned delivery corridors, vehicle types, driver employment status, home hub allocations.	~200 – 400 scheduled rider assignments daily.	Daily batch at 06:00 AM via automated SFTP. → `Fact_Rider_Route_Assignment`, `Dim_Rider`, `Dim_Shift`

8. Non-Functional Requirements (NFRs)

8.1 Data Quality & Referential Integrity (NFR-01)

- **Surrogate Key Integrity:** All foreign keys in fact tables must resolve to active surrogate keys in corresponding dimensions. Unmatched source natural keys must be assigned to default row `-1` (`"Unknown / Not Specified"`) to ensure no facts are dropped.
- **Validation Rules:** The ETL engine must enforce business rules: attempt timestamps must not precede order booking dates; delivery fee amounts must be non-negative; daily backlog parcel counts must be >= 0.

8.2 Pipeline Latency & Refresh Windows (NFR-02)

- **Nightly Batch Processing Window:** Daily ETL pipelines must execute within a strict **2-hour processing window** between 01:00 AM and 03:00 AM.
- **Snapshot Immutability:** The hub backlog snapshot must execute precisely at 23:59:59 daily, capturing true end-of-day state after sorting shifts conclude.
- **Business Readiness:** Analytical marts and dashboards must be refreshed, indexed, and available for business users no later than **06:30 AM** local time.

8.3 Query Performance & Scalability (NFR-03)

- **Interactive Dashboard Latency:** Executive summary dashboard queries must return in **< 1.5 seconds** under multi-user concurrency.

- **Historical OLAP Drill-Down:** Multi-dimensional slicing and dicing across 12 months of atomic delivery attempts (approx. 5 million rows) must execute in **< 3.0 seconds** using clustered columnstore indexing.

8.4 Security & Role-Based Access Control (NFR-04)

- **Role-Based Access:** Access is segmented: Executive Role (unrestricted access); Operations Role (operational metrics accessible, financial margins masked); Commercial Role (customer profitability accessible, rider IDs masked).
- **PII Masking:** Recipient names, personal phone numbers, and exact residential addresses must be masked or excluded from dimensional marts, maintaining only geographic routing attributes (Route Key, City, Postal Code).

8.5 Auditability & Lineage Tracking (NFR-05)

- Every fact and dimension table must include standardized audit columns: `ETL_Batch_ID`, `Source_System_Code`, `Created_Timestamp`, and `Updated_Timestamp`.

9. Assumptions, Constraints, and Risk Management

9.1 Assumptions & Constraints

- **Case Study Data Synthesis:** Datasets will be realistically generated matching the specified schema and public courier distributions to preserve proprietary commercial privacy.
- **Star Schema Constraint:** The physical architecture strictly adheres to star schemas. Snowflaking is avoided to maintain high OLAP query performance and simplified BI user navigation.
- **Pure Factless Fact Table:** `Fact_Rider_Route_Assignment` must remain strictly factless (zero numeric measure columns). Cross-process productivity metrics must be derived via BI drill-across queries.

9.2 Risk Analysis & Mitigation Matrix

Risk ID	Identified Risk Description	Severity	Likelihood	Technical & Architectural Mitigation
RSK-01	**Data Sparsity in Factless Fact:** Missing roster data from regional hubs could distort route coverage metrics.	High	Medium	Implement an automated ETL reconciliation step that cross-references actual delivery attempts against scheduled assignments, flagging unassigned routes.
RSK-02	**ETL Batch Processing Bottlenecks:** Growing delivery attempt volumes could exceed the 2-hour nightly batch window.	High	Medium	Implement date-partitioning on `Fact_Delivery_Attempt` and restrict daily ingestion strictly to delta records using database Change Data Capture (CDC).

RSK-03	**SCD Type 2 Row Explosion:** Frequent updates to non-critical rider attributes could cause excessive table bloat.	Medium	Low	Restrict SCD Type 2 tracking strictly to structurally critical attributes (Home Hub, Customer Tier). Handle minor contact updates via SCD Type 1 in-place overwrites.
RSK-04	**Ambiguous Status Code Mappings:** Operational hubs in different cities use non-standardized delivery failure codes.	Medium	High	Establish a standardized enterprise junk dimension (`Dim_Delivery_Status`) with an ETL mapping table that normalizes divergent source codes into standard categories.

10. Out-of-Scope Statement

The following technical areas are explicitly declared out-of-scope for Phase 1 to maintain project focus:

- **Real-Time IoT Telematics:** Sub-second streaming of GPS vehicle coordinates, engine diagnostics, and real-time live map tracking.
- **Statutory HR & Payroll Accounting:** Employee payroll tax withholdings, overtime penalty calculations, medical benefits, and direct bank disbursement systems.
- **Automated Sorting Warehouse PLC Systems:** Hardware-level programming for conveyor belts, robotic sorting arms, and automated guided vehicles (AGVs).
- **International Customs & Freight Forwarding:** Cross-border tariffs, customs declarations, and bonded warehouse manifest processing.

11. Requirements Traceability Matrix (RTM)

The RTM demonstrates end-to-end alignment from business requirements to functional specifications, dimensional tables, and KPIs.

BR ID	Business Requirement Summary	Linked FR	Supporting Fact Table(s)	Supporting Dimension(s)	Generated KPI(s)	Priority
BR-01	Delivery SLA & On-Time Compliance	FR-01, FR-05, FR-11	`Fact_Delivery_Attempt`	`Dim_Date`, `Dim_Service_Type`, `Dim_Hub`, `Dim_Route`	On-Time Delivery Rate (OTDR)	**Must**
BR-02	Delivery Attempt	FR-01, FR-08	`Fact_Delivery_Attempt`	`Dim_Delivery_Status`, `Dim_Customer`, `Dim_Route`	First-Attempt	**Must**

	Outcomes & Failures				Delivery Rate (FADR)	
BR-03	Daily Hub Backlog & Dwell Latency	FR-02, FR-10	Fact_Daily_Hub_Backlog	Dim_Date, Dim_Hub, Dim_Parcel_Type	Hub Backlog Ratio (HBR), Dwell Index	**Must**
BR-04	Scheduled Fleet Route Coverage	FR-03, FR-05	Fact_Rider_Route_Assignment	Dim_Date, Dim_Rider, Dim_Route, Dim_Hub, Dim_Shift	Route Coverage %, Unassigned Days	**Must**
BR-05	Rider Delivery Productivity	FR-04, FR-12	Fact_Delivery_Attempt, Fact_Rider_Route_Assignment	Dim_Date, Dim_Rider, Dim_Route, Dim_Shift	Deliveries per Rider-Hour (DPRH)	**Must**
BR-06	Route Cost-to-Serve & Net Margin	FR-01, FR-09	Fact_Delivery_Attempt	Dim_Route, Dim_Hub, Dim_Service_Type, Dim_Date	Route Net Contribution Margin (RNCM)	**Must**
BR-07	Corporate Customer Account Margin	FR-01, FR-09	Fact_Delivery_Attempt	Dim_Customer, Dim_Service_Type, Dim_Date	Customer Lifetime Margin, Volume	**Should**
BR-08	Historical Rider Transfers (SCD)	FR-06	Fact_Delivery_Attempt, Fact_Rider_Route_Assignment	Dim_Rider (SCD Type 2)	Historical Rider Attribution	**Should**
BR-09	Customer Tier Migrations (SCD)	FR-07	Fact_Delivery_Attempt	Dim_Customer (SCD Type 2)	Tier Migration Trends, Churn	**Should**
BR-10	Parcel Volumetric Capacity Impact	FR-01, FR-02	Fact_Delivery_Attempt, Fact_Daily_Hub_Backlog	Dim_Parcel_Type, Dim_Hub, Dim_Route	Volumetric Capacity Index	**Could**

12. Acceptance Criteria & Business Verification

Acceptance criteria define how business stakeholders will formally verify the completion and accuracy of each **Must-Have** requirement.

Requirement	Verification Scenario	Test Procedure	Acceptance Threshold
BR-01 (OTDR)	Validate on-time delivery rate calculations across delivery tiers.	Aggregate `Fact_Delivery_Attempt` for a control month. Compare attempts where `Attempt_Timestamp <= SLA_Deadline` against total successful deliveries.	OTDR reconciles 100% exactly with operational control totals.
BR-02 (FADR)	Verify that multi-attempt shipments record independent rows per attempt.	Query a shipment with 3 failed attempts and 1 success. Confirm 4 distinct rows exist in `Fact_Delivery_Attempt` with sequence numbers 1 to 4 and correct status keys.	Zero attempt rows dropped; FADR accurately filters strictly for sequence 1 successes.
BR-03 (Hub Backlog)	Verify semi-additive aggregation behavior of hub inventory balances.	Query `Fact_Daily_Hub_Backlog` across Hubs A and B for Monday and Tuesday. Sum across hubs for Monday (valid); average across dates (valid); sum across dates (invalid).	Semi-additive aggregation rules enforced correctly in OLAP semantic layer.
BR-04 (Factless Fleet)	Confirm that `Fact_Rider_Route_Assignment` contains zero numeric measures.	Inspect physical table schema. Verify table consists purely of 5 foreign keys (Date, Rider, Route, Hub, Shift). Query for unassigned routes via outer join.	Schema validation passes with 0 numeric measure columns; identifies unassigned routes.

BR-05 (Productivity)	Validate calculation of Deliveries per Rider-Hour via drill-across query.	Execute multi-fact query joining `Fact_Delivery_Attempt` with `Fact_Rider_Route_Assignment` and `Dim_Shift` along conformed Date and Rider keys.	Computed DPRH matches manual verification (e.g., 28 deliveries / 8 hours = 3.5).
BR-06 (Route Margin)	Validate additive financial roll-up of route profitability.	For Route RT-101, calculate: `Sum(Delivery_Fee_Amount) - Sum(Estimated_Labor_Cost + Estimated_Fuel_Cost)`. Compare with raw source billing.	Calculated net margin reconciles within $0.01 of source accounting controls.

13. Viva Defense Guide: Key Design Decisions in this BRD

Design Decision 1: Pure Factless Fact for Rider Route Assignment (Fact_Rider_Route_Assignment)

Examiner Question: *"Why did you choose a factless fact for rider assignments instead of storing delivered package counts and hours worked directly in that table?"*

Architectural Defense:
*"According to Ralph Kimball, combining distinct business processes into a single fact table corrupts the grain. A workforce assignment is a **scheduling event** established prior to delivery dispatch, whereas parcel deliveries are **operational transactions** occurring continuously throughout the day. If we stored delivered package counts inside the assignment table, we would be forced to pre-aggregate delivery transactions, discarding critical atomic details such as individual attempt timestamps, GPS coordinates, and failure reason codes. Furthermore, storing hours worked directly in the assignment table would violate the factless fact pattern. Instead, we model `Fact_Rider_Route_Assignment` as a pure factless fact containing foreign keys only (`Date_Key`, `Rider_Key`, `Route_Key`, `Hub_Key`, `Shift_Key`). To compute **Deliveries per Rider-Hour**, the BI semantic layer executes a clean **multi-fact drill-across** query, combining the assignment fact with `Fact_Delivery_Attempt` along conformed Date and Rider dimensions, and drawing standard working hours from `Dim_Shift`. This preserves atomic granularity and adheres strictly to Kimball's Enterprise Bus Architecture."*

Design Decision 2: Periodic Snapshot Fact for Hub Backlog (Fact_Daily_Hub_Backlog)

Examiner Question: *"Why do we need a separate periodic snapshot fact table for hub backlogs when all parcel scan events are already recorded?"*

Architectural Defense:
*"In logistics, inventory is a **state**, not an event. While it is theoretically possible to derive end-of-day parcel*

backlog by running a cumulative running-sum query (inbound scans minus outbound scans) from raw transaction scan logs, doing so across millions of historical scan rows in a multi-year data warehouse causes severe query performance degradation. Answering a simple question such as 'What was Hub A's backlog on March 15th?' would require scanning every historical scan event up to that date.
*Ralph Kimball recommends the **Periodic Snapshot Fact** pattern for inventory balances. By taking an automated snapshot at 23:59:59 daily, we freeze and preserve the end-of-day backlog count, average dwell time, and aging profile. This allows Hub Managers to analyze 12-month backlog trends and dwell-time spikes in sub-second query times without placing compute overhead on transactional scan logs."*

Design Decision 3: Delivery Status Junk Dimension (Dim_Delivery_Status)

Examiner Question: *"Why did you create a junk dimension instead of keeping status flags directly in the fact table or creating separate tables for each status flag?"*

Architectural Defense:
"Courier dispatch involves multiple low-cardinality flags and descriptors: `Attempt_Status` (Delivered, Failed, Returned), `Failure_Reason_Code` (Customer Unavailable, Incorrect Address, Gate Locked, Weather Exception), and `Signature_Required_Flag` (Yes, No). If we store these text attributes directly inside `Fact_Delivery_Attempt`, the fact table widens significantly, which increases disk I/O and degrades columnar compression across millions of transaction rows. Conversely, creating three separate dimension tables creates unnecessary snowflake clutter and forces the database engine to perform three separate joins for basic status filtering.
*Kimball's **Junk Dimension** resolves this by pre-computing all distinct combinations of these low-cardinality indicators into a single compact dimension (`Dim_Delivery_Status`). This reduces fact table width to a single foreign key, maximizes data compression, and simplifies BI report creation."*